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Evidence-backed exploratory analytics, KPI summary, and strategic findings

| DATASET | RECORDS / SCOPE | REPORTING PERIOD | GENERATED DATE |
|---|---|---|---|
| sample_sales.csv | 10 rows (8 cols) | 2024-01-10 to 2024-01-30 | 18/10/2026 |

## Executive Summary

**Column 'quantity' contains 1 outlier values (10% of records) falling outside the IQR threshold interval [-0.5, 3.5]. Minimum is 1, maximum is 4, and median is 1.5.**

- Outlier Detection in 'quantity': Column 'quantity' contains 1 outlier values (10% of records) falling outside the IQR threshold interval [-0.5, 3.5]. Minimum is 1, maximum is 4, and median is 1.5.
- Outlier Detection in 'unit_price': Column 'unit_price' contains 1 outlier values (11.11% of records) falling outside the IQR threshold interval [-600.235, 1160.125]. Minimum is 25, maximum is 1250, and median is 120.
- Temporal Trend: INCREASING (+150.01% over period): Activity tracked along 'transaction_date' spans 20 days (2024-01-10 to 2024-01-30) across 10 daily periods. The net period-over-period trajectory changed by 150.01% with a 'increasing' pattern. Unusual volatility was observed in periods: [2024-01-30].
- Concentration in 'product_category': 'Electronics' Dominance: The top category 'Electronics' in column 'product_category' represents 40% of all valid records (4 of 10 non-null rows). There are 4 distinct categories in total.
- Correlation Identified: 'unit_price' vs 'quantity' (r = -0.58): A statistically moderate negative linear association (Pearson r = -0.58, sample size n = 9) exists between 'unit_price' and 'quantity'.

**Data Trust Rating:** HIGH — Evaluated over 10 rows. Data completeness score is 98.75%. 2 quality notices identified.

## Executive Key Performance Indicators

TOTAL RECORDS

**10**

Total validated rows analyzed across the dataset.

COMPLETENESS SCORE

**98.8%**

Ratio of non-null cells to total cells across all rows and columns.

MEAN UNIT_PRICE

**$346.51**

Arithmetic mean computed across 9 non-null values.

MEAN QUANTITY

**1.80**

Arithmetic mean computed across 10 non-null values.

PERIOD GROWTH RATE

**+150.0%**

+150.0% vs 2024-01-10 to 2024-01-30

Observed trend direction: increasing across 10 temporal buckets.

## Data Quality & Completeness Scorecard

Overall Score: **98.8%** Status: **GOOD** Columns with Nulls: **1 / 8**

| Column | Data Type | Completeness % | Null Count | Identified Issues | Status |
|---|---|---|---|---|---|
| **unit_price** | NUMERIC | 90.0% | 1 | Column 'unit_price' exhibits 1 outliers (11.11% of values outside [-600.235, 1160.125]). | WARNING |
| **quantity** | NUMERIC | 100.0% | 0 | Column 'quantity' exhibits 1 outliers (10% of values outside [-0.5, 3.5]). | INFO |
| **transaction_id** | CATEGORICAL | 100.0% | 0 | None detected | INFO |
| **customer_id** | CATEGORICAL | 100.0% | 0 | None detected | INFO |
| **product_category** | CATEGORICAL | 100.0% | 0 | None detected | INFO |
| **discount_applied** | CATEGORICAL | 100.0% | 0 | None detected | INFO |
| **payment_status** | CATEGORICAL | 100.0% | 0 | None detected | INFO |

*Scorecard covers 100% of rows analyzed across all attributes. Null percentages reflect unpopulated cells.*

## Ranked Strategic Findings & Recommendations

### #1 — Outlier Detection in 'quantity'

HIGH PRIORITY

**OBSERVED FACT (EVIDENCE-BACKED)**

Column 'quantity' contains 1 outlier values (10% of records) falling outside the IQR threshold interval [-0.5, 3.5]. Minimum is 1, maximum is 4, and median is 1.5.

**EXPLORATORY HYPOTHESIS**

Extreme values in 'quantity' could reflect high-value premium transactions, bulk institutional orders, or manual data entry anomalies.

**RECOMMENDED ACTION**

Conduct segmentation analysis excluding these 1 extreme records to compare baseline core behavior against premium outliers.

**Limitations / Caveats:** IQR boundaries assume unimodal distributions; highly multi-modal data may require domain-specific thresholds.

### #2 — Outlier Detection in 'unit_price'

HIGH PRIORITY

**OBSERVED FACT (EVIDENCE-BACKED)**

Column 'unit_price' contains 1 outlier values (11.11% of records) falling outside the IQR threshold interval [-600.235, 1160.125]. Minimum is 25, maximum is 1250, and median is 120.

**EXPLORATORY HYPOTHESIS**

Extreme values in 'unit_price' could reflect high-value premium transactions, bulk institutional orders, or manual data entry anomalies.

**RECOMMENDED ACTION**

Conduct segmentation analysis excluding these 1 extreme records to compare baseline core behavior against premium outliers.

**Limitations / Caveats:** IQR boundaries assume unimodal distributions; highly multi-modal data may require domain-specific thresholds.

### #3 — Temporal Trend: INCREASING (+150.01% over period)

HIGH PRIORITY

**OBSERVED FACT (EVIDENCE-BACKED)**

Activity tracked along 'transaction_date' spans 20 days (2024-01-10 to 2024-01-30) across 10 daily periods. The net period-over-period trajectory changed by 150.01% with a 'increasing' pattern. Unusual volatility was observed in periods: [2024-01-30].

**EXPLORATORY HYPOTHESIS**

Positive growth over this window may correspond to seasonality, marketing acquisition surges, or expanding customer activity.

**RECOMMENDED ACTION**

Forecast near-term capacity and supply chain requirements assuming continuation of current trajectory, validating against prior seasonal cycles.

**Limitations / Caveats:** Evaluates aggregate volume change between start and end intervals; external seasonal factors require multi-year historical data to establish cyclicality.

### #4 — Concentration in 'product_category': 'Electronics' Dominance

MEDIUM PRIORITY

**OBSERVED FACT (EVIDENCE-BACKED)**

The top category 'Electronics' in column 'product_category' represents 40% of all valid records (4 of 10 non-null rows). There are 4 distinct categories in total.

**EXPLORATORY HYPOTHESIS**

Strong preference or volume concentration in 'Electronics' suggests either product-market fit or an over-reliance on a single segment.

**RECOMMENDED ACTION**

Evaluate operational dependencies on 'Electronics'. Determine if secondary segments ('Apparel', 'Home & Garden') can be expanded through targeted campaigns.

**Limitations / Caveats:** Category distribution reflects transaction frequency, which may differ from dollar-weighted revenue volume.

### #5 — Correlation Identified: 'unit_price' vs 'quantity' (r = -0.58)

MEDIUM PRIORITY

**OBSERVED FACT (EVIDENCE-BACKED)**

A statistically moderate negative linear association (Pearson r = -0.58, sample size n = 9) exists between 'unit_price' and 'quantity'.

**EXPLORATORY HYPOTHESIS**

Higher values in 'unit_price' coincide with lower values in 'quantity', potentially indicating an inverse trade-off or discounting effect.

**RECOMMENDED ACTION**

Analyze paired scatter plots and residual distributions. Prior to making operational policy changes, execute a controlled A/B test or multivariate regression to verify whether an actual causal mechanism exists.

**Limitations / Caveats:** CRITICAL: Correlation does NOT demonstrate or imply causation. Unmeasured confounders or external factors may account for the relationship. Linear correlation assesses only linear co-movement and does not capture non-linear or threshold dependencies.

### #6 — Operational Frequency: 'discount_applied' Baseline (false)

LOW PRIORITY

**OBSERVED FACT (EVIDENCE-BACKED)**

Operational baseline in 'discount_applied': 'false' represents 60% of valid records (6 of 10 non-null rows).

**EXPLORATORY HYPOTHESIS**

Standard operational workflow baseline where the predominant outcome is 'false'.

**RECOMMENDED ACTION**

Monitor 'discount_applied' for shifts in status rates, but treat this as a descriptive baseline rather than an executive revenue driver unless correlated with revenue loss or operational bottlenecks.

**Limitations / Caveats:** Status frequencies reflect operational states and are not primary drivers of commercial strategy.

### #7 — Operational Frequency: 'payment_status' Baseline (Completed)

LOW PRIORITY

**OBSERVED FACT (EVIDENCE-BACKED)**

Operational baseline in 'payment_status': 'Completed' represents 80% of valid records (8 of 10 non-null rows).

**EXPLORATORY HYPOTHESIS**

Standard operational workflow baseline where the predominant outcome is 'Completed'.

**RECOMMENDED ACTION**

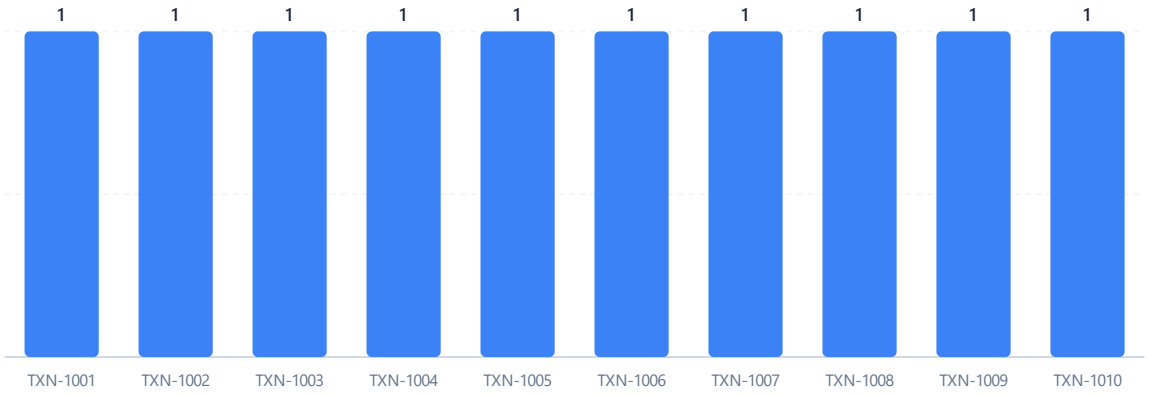
Monitor 'payment_status' for shifts in status rates, but treat this as a descriptive baseline rather than an executive revenue driver unless correlated with revenue loss or operational bottlenecks.

**Limitations / Caveats:** Status frequencies reflect operational states and are not primary drivers of commercial strategy.

## Analytical Visualizations

### Distribution by transaction_id

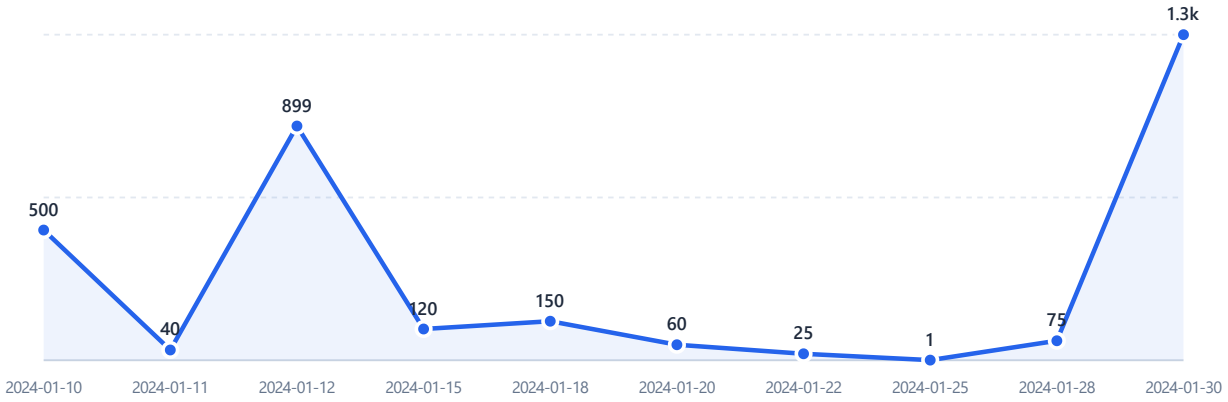
Frequency distribution across top 10 distinct categories




*Shows volume concentration for column 'transaction_id'. Mode is 'TXN-1001' representing 10% of records. Note: counts represent volume, not revenue unless explicitly aggregated.*

### Temporal Trend: transaction_date

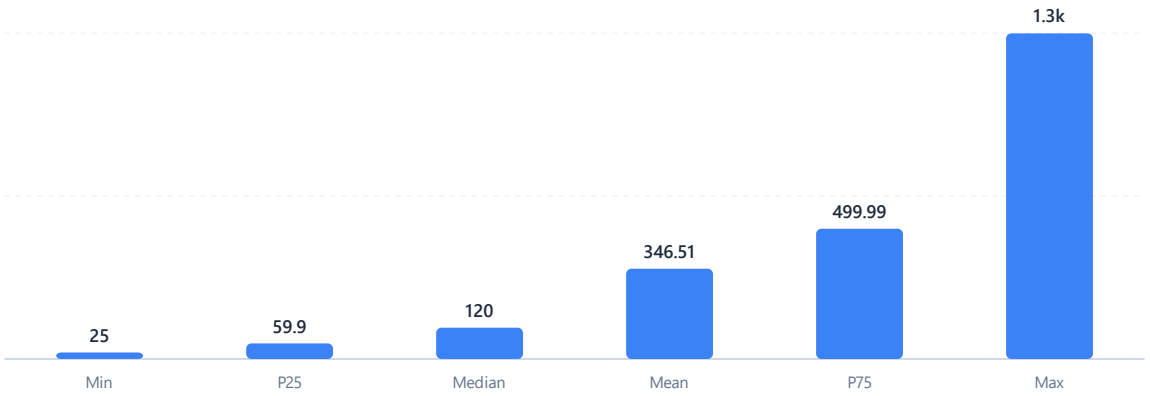
DAILY aggregation from 2024-01-10 to 2024-01-30




*Temporal trend across 10 periods. Overall observed direction is 'increasing' with calculated growth rate of 150.01%.*

### Distribution Profile: unit_price

Five-number summary and mean computed via DuckDB




*Distribution profile for 'unit_price'. Median = $120, Mean = $346.51, IQR = 440.09. 1 outliers identified (11.1%).*

## Detailed Tabular Summaries

### Numeric Attributes Summary Statistics

| Column | Count | Null % | Mean | Median | Std Dev | Min | Max | Outliers |
|---|---|---|---|---|---|---|---|---|
| unit_price | 9 | 10.0% | $346.51 | $120 | 445.59 | $25 | $1,250 | 1 (11.1%) |
| quantity | 10 | 0.0% | 1.80 | 1.50 | 1.03 | 1 | 4 | 1 (10.0%) |

*Full-dataset summary metrics computed via DuckDB analytical SQL engine. Outliers computed using Tukey's 1.5 IQR threshold.*

### Categorical Attributes Summary

| Column | Distinct Values | Null % | Mode (Top Value) | Mode Count | Top Value Share |
|---|---|---|---|---|---|
| transaction_id | 10 | 0.0% | TXN-1001 | 1 | 10.0% |
| customer_id | 9 | 0.0% | CUST-201 | 2 | 20.0% |
| product_category | 4 | 0.0% | Electronics | 4 | 40.0% |
| discount_applied | 2 | 0.0% | false | 6 | 60.0% |
| payment_status | 3 | 0.0% | Completed | 8 | 80.0% |

*Categorical distribution summary. Unique ratios and mode concentrations computed across entire authorized dataset.*

## Statistical Distribution & Association Analysis

**CRITICAL STATISTICAL NOTICE: Correlation measures mathematical association between variables in this dataset. It does not establish cause-and-effect relationships or directionality. Business decisions should not assume that changing one variable will produce changes in correlated variables without controlled experimentation or causal identification.**

**Temporal Trend Assessment:** Time-series assessment over 'transaction_date' from 2024-01-10 to 2024-01-30 demonstrated a increasing direction across 10 reporting intervals.

### Pairwise Pearson Correlation Analysis

| Variable A | Variable B | Pearson r | Sample Size (N) | Statistical Interpretation |
|---|---|---|---|---|
| **unit_price** | **quantity** | **-0.584** | 9 | Observed Pearson correlation coefficient r = -0.584 indicates a moderate negative relationship across 9 paired observations. |

### Identified Statistical Outliers (Tukey IQR Criterion)

| Column | Outlier Count | % of Records | Acceptable Bounds [Lower, Upper] | Business Implication |
|---|---|---|---|---|
| **unit_price** | 1 | 11.1% | [-600.24, 1160.13] | 1 data point(s) exceed statistical bounds [-600.24, 1160.13]. Review for recording anomalies or high-impact tail events. |
| **quantity** | 1 | 10.0% | [-0.50, 3.50] | 1 data point(s) exceed statistical bounds [-0.50, 3.50]. Review for recording anomalies or high-impact tail events. |

## Methodology, Limitations & Provenance

**Analytical Engine:** DuckDB Embedded SQL & Statistical Profiler

**Data Profiling Method:** Exhaustive streaming scan and exact SQL aggregation over complete dataset rows

**Source File:** sample_sales.csv

**Analysis Execution Time:** 131 ms

### Statistical Appendix & Metric Definitions

**Interquartile Range (IQR) Outlier Rule**

`Lower Bound = Q1 - (1.5 × IQR); Upper Bound = Q3 + (1.5 × IQR)`

Identifies values that fall beyond 1.5 times the spread of the middle 50% of the distribution. Established by John Tukey for robust outlier detection resistant to extreme values.

**Pearson Correlation Coefficient (r)**

$r = \Sigma((x - \bar{x})(y - \bar{y})) / (\sqrt{\Sigma(x - \bar{x})^2} \times \sqrt{\Sigma(y - \bar{y})^2})$

Measures linear relationship strength between two continuous variables, bounded between -1.0 (perfect negative) and +1.0 (perfect positive). Does NOT imply causal relationship.

**Completeness Score (%)**

`(1 - (Total Null Cells / Total Cell Count)) × 100%`

Measures overall dataset fill rate across all recorded rows and columns. Scores above 95% indicate high data completeness.

**Categorical Mode Concentration (%)**

`(Frequency of Most Frequent Value / Total Non-Null Count) × 100%`

Indicates the dominance of the primary category. High concentration (>60%) flags potential class imbalance or monopolistic segments.